Estimating the Hessian and Inverse Hessian via Secant Equations and BFGS

Goal

- Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function.
- Suppose we know two iterates: $x^{(k)},\ x^{(k+1)} \in \mathbb{R}^n$ and their gradients: $\nabla f(x^{(k)}), \quad \nabla f(x^{(k+1)}) \in \mathbb{R}^n$.
- We want to estimate: $\nabla^2 f(x^{(k+1)})$, or more practically, its inverse $\left(\nabla^2 f(x^{(k+1)})\right)^{-1}$.
- **Exact Newton method:** $x^{(k+1)} = x^{(k)} - \left(\nabla^2 f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$
- **Quasi-Newton idea:** replace the true inverse Hessian by an approximation $H^{(k)}$: $p^{(k)} = -H^{(k)} \nabla f(x^{(k)})$
- Then update: $x^{(k+1)} = x^{(k)} + \alpha^{(k)} p^{(k)}$, where $\alpha^{(k)} > 0$ is a step length. This is why inverse Hessian approximations are often preferred in practice. This requires computing and inverting the true Hessian.

Important Notation

Standard quasi-Newton notation:

$$B^{(k)} \approx \nabla^2 f(x^{(k)})$$

$$H^{(k)} \approx \left(\nabla^2 f(x^{(k)})\right)^{-1}$$

Interpretation:

- $B^{(k)}$: approximation of the Hessian
- $H^{(k)}$: approximation of the inverse Hessian

Memory trick:

- B maps **step** $\rightarrow$ **gradient change**
- H maps **gradient change** $\rightarrow$ **step**

Why the Hessian Matters

- The Hessian is the Jacobian of the gradient: $\nabla^2 f(x) = \frac{\partial(\nabla f(x))}{\partial x}$
- Therefore, it describes how the gradient changes as the input changes.
- In the one-dimensional secant method, the derivative is approximated using two function evaluations: $\boxed{f'(x^{(k)}) \approx \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}}$
- Let $f : \mathbb{R}^n \to \mathbb{R}$ and consider two points with their gradient information: $\left(x^{(k-1)}, \nabla f(x^{(k-1)})\right), \left(x^{(k)}, \nabla f(x^{(k)})\right)$. How can we estimate $\nabla^2 f(x^{(k)})$?
- The Hessian describes how the gradient changes with respect to the input. Hence, using the secant idea, $B^{(k)} \approx \nabla^2 f(x^{(k)}) \approx \frac{\nabla f(x^{(k)}) - \nabla f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$.
- Let the **step vector** $\delta^{(k)} = x^{(k)} - x^{(k-1)}$, and the **gradient-difference vector** $y^{(k)} = \nabla f(x^{(k)}) - \nabla f(x^{(k-1)})$.
- $\boxed{B^{(k)} \approx \frac{y^{(k)}}{\delta^{(k)}}}$ and $\boxed{H^{(k)} \approx \frac{\delta^{(k)}}{y^{(k)}}}$

1D Case: Secant Approximation

- When $n = 1$, all quantities are scalars.
- Then: $\delta^{(k)} = x^{(k)} - x^{(k-1)} \in \mathbb{R}, \qquad y^{(k)} = f'(x^{(k)}) - f'(x^{(k-1)}) \in \mathbb{R}.$
- So the Hessian (second derivative) is approximated by:

$$B^{(k)} = f''(x^{(k)}) \approx \frac{y^{(k)}}{\delta^{(k)}}.$$

- And the inverse Hessian is approximated by:

$$H^{(k)} = \left(f''(x^{(k+1)})\right)^{-1} \approx \frac{\delta^{(k)}}{y^{(k)}}, \qquad \text{if } y^{(k)} \neq 0.$$

In 1D, division is valid because everything is scalar.

- This pair $(\delta^{(k)}, y^{(k)})$ is called the **secant pair**. because it captures how the gradient changes between two nearby iterates, providing an average curvature estimate over that interval, just as a secant line approximates slope using two points. local curvature information.

Multi-Dimensional Case: No Direct Division

- When $n > 1$: $\delta^{(k)} \in \mathbb{R}^n, \qquad y^{(k)} \in \mathbb{R}^n, \qquad \nabla^2 f(x^{(k)}) \in \mathbb{R}^{n \times n}.$
- In this case, the expression $\frac{y^{(k)}}{\delta^{(k)}}$ is not defined as ordinary vector division.
- Therefore, in multiple dimensions, we do **not** write

$$B^{(k)} = \nabla^2 f(x^{(k)}) \approx \frac{y^{(k)}}{\delta^{(k)}}.$$

or

$$H^{(k)} = \nabla^2 f(x^{(k)})^{-1} \approx \frac{\delta^{(k)}}{y^{(k)}}.$$

- Instead, we use a **matrix equation**: the secant equation.
  - $B^{(k+1)}\delta^{(k)} = y^{(k)}$. B maps **step** $\rightarrow$ **gradient change**
  - then the inverse secant equation is: $H^{(k+1)}y^{(k)} = \delta^{(k)}$. H maps **gradient change** $\rightarrow$ **step**

Why One Secant Equation Does Not Determine the Whole Hessian

- Suppose we want to recover the Hessian approximation $B^{(k+1)} \in \mathbb{R}^{n \times n}$ from the secant equation

$$B^{(k+1)}\delta^{(k)} = y^{(k)}.$$

- Here, $\delta^{(k)}$, $y^{(k)} \in \mathbb{R}^n$ are **known vectors**, while $B^{(k+1)} \in \mathbb{R}^{n \times n}$ is the **unknown Hessian approximation**.
- A general $n \times n$ matrix contains $\underbrace{n^2}_{\text{unknown entries}}$ unknown elements.
- The secant condition $B^{(k+1)}\delta^{(k)} = y^{(k)}$ is a vector equation in $\mathbb{R}^n$, and therefore provides only $\underbrace{n}_{\text{scalar equations}}$ independent equations.
- Moreover, since a Hessian is symmetric $B^{(k+1)} = \left(B^{(k+1)}\right)^T$. We only need to determine $\underbrace{\frac{n(n+1)}{2}}_{\text{independent unknowns}}$ remain.

- For $n > 1$, $n < \frac{n(n+1)}{2}$.
- This is because $\frac{n(n+1)}{2} - n = \frac{n(n-1)}{2} > 0, \qquad n > 1.$
- Hence, the symmetric Hessian contains $\frac{n(n+1)}{2}$ independent unknowns, whereas the secant equation $B^{(k+1)}\delta^{(k)} = y^{(k)}$ provides only n scalar equations.
- Consider the two-dimensional case, $n = 2$. The symmetric Hessian approximation is $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \end{bmatrix}$, so it has **three independent unknowns**: $b_{11}, \quad b_{12}, \quad b_{22}$.
- The secant condition $B^{(k)}\delta^{(k)} = y^{(k)}$ gives only **two scalar equations**:
$$b_{11}\delta_1^{(k)} + b_{12}\delta_2^{(k)} = y_1^{(k)},$$
$$b_{12}\delta_1^{(k)} + b_{22}\delta_2^{(k)} = y_2^{(k)}.$$
- Therefore, $\underbrace{2}_{\text{equations}} < \underbrace{3}_{\text{unknowns}}$.
- Hence, one observed step $\delta^{(k)}$ and its corresponding gradient change $y^{(k)}$ **cannot uniquely determine all entries of the Hessian approximation** B.

- Consequently, the secant condition gives an **underdetermined system**. In general, infinitely many symmetric matrices $B^{(k+1)}$ can satisfy $B^{(k+1)}\delta^{(k)} = y^{(k)}$.
- Hence, the secant condition alone cannot uniquely determine the Hessian approximation.
- Therefore, additional conditions are required to select a **unique Hessian approximation** from the infinitely many matrices satisfying the secant condition.
- BFGS selects the new approximation $B^{(k+1)}$ by requiring it to:
  - satisfy the secant condition;
  - remain symmetric;
  - preserve positive definiteness when $(\delta^{(k)})^T y^{(k)} > 0$; and
  - change the previous approximation $B^{(k)}$ as little as necessary.
- These requirements lead to the specific **rank-two BFGS update** used to construct $B^{(k+1)}$.

Minimal-Change Interpretation

A useful interpretation is:

- Among all symmetric matrices satisfying the new secant information, choose the one that stays **closest to the previous Hessian approximation**.
- For the Hessian approximation:

$$\begin{aligned} \arg\min_{B^{(k+1)}} \quad & \|B^{(k+1)} - B^{(k)}\|_F^2 \\ \text{subject to} \quad & (B^{(k+1)})^T = B^{(k+1)}, \\ & B^{(k+1)}\delta^{(k)} = y^{(k)}. \end{aligned}$$

- Here,

$$\|A\|_F = \sqrt{\text{tr}(A^T A)}$$

is the Frobenius norm, which measures the overall magnitude of the matrix difference.

Minimal-Change Interpretation

A useful interpretation is:
Among all matrices satisfying the new secant information, choose the one that stays **closest** to the previous approximation.
For the inverse Hessian approximation:

$$\begin{aligned} \arg\min_{H^{(k+1)}} \quad & \|H^{(k+1)} - H^{(k)}\|_F^2 \\ \text{subject to} \quad & \left(H^{(k+1)}\right)^T = H^{(k+1)}, \\ & H^{(k+1)} y^{(k)} = \delta^{(k)}. \end{aligned}$$

Here

$$\|A\|_F = \sqrt{\text{tr}(A^T A)}$$

is the Frobenius norm.

BFGS: Hessian Update (Direct Form)

For a direct Hessian approximation,

$$B^{(k)} \approx \nabla^2 f(x^{(k)}),$$

the BFGS Hessian update is

$$B^{(k+1)} = B^{(k)} - \frac{B^{(k)}\delta^{(k)}\left(\delta^{(k)}\right)^T B^{(k)}}{\left(\delta^{(k)}\right)^T B^{(k)}\delta^{(k)}} + \frac{y^{(k)}\left(y^{(k)}\right)^T}{\left(y^{(k)}\right)^T \delta^{(k)}}.$$

where

$$\delta^{(k)} = x^{(k)} - x^{(k-1)}, \qquad y^{(k)} = \nabla f(x^{(k)}) - \nabla f(x^{(k-1)}).$$

The update satisfies the secant condition

$$\boxed{B^{(k+1)}\delta^{(k)} = y^{(k)}}.$$

BFGS: Inverse Hessian Update

- For the inverse Hessian approximation, $H^{(k)} \approx \left(\nabla^2 f(x^{(k)})\right)^{-1}$, The BFGS inverse-Hessian update is

$$\boxed{H^{(k+1)} = \left(I - \rho^{(k)} \delta^{(k)} \left(y^{(k)}\right)^T\right) H^{(k)} \left(I - \rho^{(k)} y^{(k)} \left(\delta^{(k)}\right)^T\right) + \rho^{(k)} \delta^{(k)} \left(\delta^{(k)}\right)^T}$$

- where

$$\rho^{(k)} = \frac{1}{\left(y^{(k)}\right)^T \delta^{(k)}} \quad \text{is the curvature condition,}$$

  and $I \in \mathbb{R}^{n \times n}$ is the identity matrix.

- It update satisfies the inverse secant condition

$$\boxed{H^{(k+1)} y^{(k)} = \delta^{(k)}}.$$

Curvature Condition

- **Curvature condition:**

$$\left(y^{(k)}\right)^T \delta^{(k)} > 0 \quad \Longrightarrow \quad \rho^{(k)} = \frac{1}{\left(y^{(k)}\right)^T \delta^{(k)}} > 0$$

- **Where:** $\delta^{(k)} = x^{(k)} - x^{(k-1)}$: the step
- $y^{(k)} = \nabla f(x^{(k)}) - \nabla f(x^{(k-1)})$: change in gradient along that step
- **Interpretation:** $\left(y^{(k)}\right)^T \delta^{(k)} > 0$ means the change in gradient points roughly in the same direction as the step. So the function is curving upward (locally convex) along that direction, like a bowl shape. This helps keep the approximate Hessian (or inverse Hessian) **positive definite**, which guarantees **descent directions**. Why?
- BFGS uses $H^{(k)}$ to define $p^{(k)} = -H^{(k)} \nabla f(x^{(k)})$.
- If $H^{(k)} \succ 0$ and $\nabla f(x^{(k)}) \neq 0$, then directional derivative at $p^{(k)}$

$$\nabla f(x^{(k)})^T p^{(k)} = -\nabla f(x^{(k)})^T H^{(k)} \nabla f(x^{(k)}) < 0$$

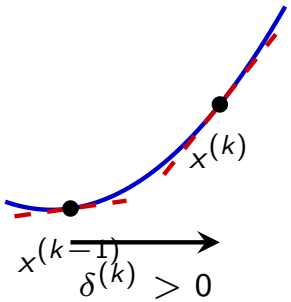
- Because $H^{(k)} \succ 0$. Hence, $p^{(k)}$ is a **descent direction** at

Geometric Interpretation of Curvature Condition

Assuming a forward step $\delta^{(k)} = x^{(k)} - x^{(k-1)} > 0$:

$(y^{(k)})^T \delta^{(k)} > 0$

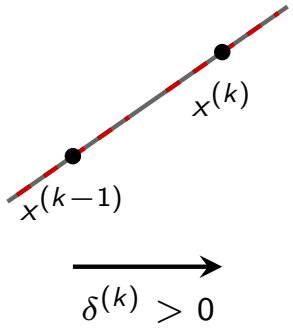
Locally Convex (Bowl)



Slope increases:
$\nabla f(x^{(k)}) > \nabla f(x^{(k-1)})$
$\implies y^{(k)} > 0$

$(y^{(k)})^T \delta^{(k)} = 0$

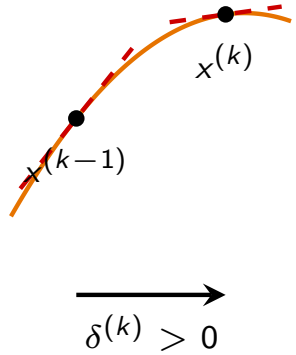
Zero Curvature (Linear)



Slope unchanged:
$\nabla f(x^{(k)}) = \nabla f(x^{(k-1)})$
$\implies y^{(k)} = 0$

$(y^{(k)})^T \delta^{(k)} < 0$

Locally Concave (Cap)



Slope decreases:
$\nabla f(x^{(k)}) < \nabla f(x^{(k-1)})$
$\implies y^{(k)} < 0$

Implication for Quasi-Newton Updates (BFGS/SR1): Only the condition $(y^{(k)})^T \delta^{(k)} > 0$ guarantees that the updated inverse Hessian approximation $B^{(k+1)} \succ 0$ stays positive definite.

Curvature Condition from First-Order Convexity

First-order condition of convexity:
If f is convex, then for any two points $x^{(k-1)}$ and $x^{(k)}$,

$$f(x^{(k-1)}) \geq f(x^{(k)}) + \nabla f(x^{(k)})^T (x^{(k-1)} - x^{(k)}) \tag{1}$$

Swapping $x^{(k-1)}$ and $x^{(k)}$, we get:

$$f(x^{(k)}) \geq f(x^{(k-1)}) + \nabla f(x^{(k-1)})^T (x^{(k)} - x^{(k-1)}) \tag{2}$$

Adding the Convexity Inequalities

Add equations (1) and (2):

$$f(x^{(k-1)}) + f(x^{(k)}) \geq f(x^{(k)}) + f(x^{(k-1)}) + \nabla f(x^{(k)})^T (x^{(k-1)} - x^{(k)}) + \nabla f(x^{(k-1)})^T (x^{(k)} - x^{(k-1)})$$

Cancel the common terms $f(x^{(k-1)}) + f(x^{(k)})$:

$$0 \geq \nabla f(x^{(k)})^T (x^{(k-1)} - x^{(k)}) + \nabla f(x^{(k-1)})^T (x^{(k)} - x^{(k-1)})$$

Rearranging:

$$0 \geq -\left(\nabla f(x^{(k)}) - \nabla f(x^{(k-1)})\right)^T \left(x^{(k)} - x^{(k-1)}\right)$$

Hence,

$$\left(\nabla f(x^{(k)}) - \nabla f(x^{(k-1)})\right)^T \left(x^{(k)} - x^{(k-1)}\right) \geq 0$$

Conclusion from Convexity

Using

$$y^{(k)} = \nabla f(x^{(k)}) - \nabla f(x^{(k-1)}), \qquad \delta^{(k)} = x^{(k)} - x^{(k-1)},$$

we obtain

$$\left(y^{(k)}\right)^T \delta^{(k)} \geq 0.$$

Important remark:

Convexity guarantees

$$\left(y^{(k)}\right)^T \delta^{(k)} \geq 0,$$

but not automatically

$$\left(y^{(k)}\right)^T \delta^{(k)} > 0.$$

Strict positivity usually comes from:

- strict convexity, or
- a nonzero step with positive curvature, or
- line-search conditions (e.g. Wolfe conditions)

Case 1: If $\left(y^{(k)}\right)^T \delta^{(k)} = 0$

If

$$\left(y^{(k)}\right)^T \delta^{(k)} = 0,$$

then the observed curvature along the step is zero.
This means $y^{(k)}$ is orthogonal to $\delta^{(k)}$, suggesting a **flat region** with no detectable curvature along that direction.
Then

$$\rho^{(k)} = \frac{1}{\left(y^{(k)}\right)^T \delta^{(k)}}$$

is undefined.
So the BFGS update cannot be safely applied.
Practical consequence:

- updating the Hessian may introduce numerical instability
- therefore, the update is usually skipped

Case 2: If $\left(y^{(k)}\right)^T \delta^{(k)} < 0$

If

$$\left(y^{(k)}\right)^T \delta^{(k)} < 0,$$

then the gradient change $y^{(k)}$ points opposite to the step $\delta^{(k)}$.
This suggests:

- concave curvature, or
- saddle-like behavior along that direction

If this is used directly in BFGS, the approximate Hessian (or inverse Hessian) may become **indefinite**.
Then we may lose:

- positive definiteness
- the descent property
- numerical stability

Practical consequence:
To preserve positive definiteness, BFGS-type algorithms usually:

- skip the update, or
- use a damped / modified update

BFGS Algorithm (Pseudocode)

Require: Differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, gradient ∇f, initial point $x^{(0)}$, tolerance $\varepsilon > 0$
Ensure: Approximate local minimizer x^*

$k \leftarrow 0$
Compute $g^{(0)} \leftarrow \nabla f(x^{(0)})$
Set $H^{(0)} \leftarrow I$
while $\|g^{(k)}\| > \varepsilon$ **do**
    Compute search direction: $p^{(k)} \leftarrow -H^{(k)} g^{(k)}$
    Choose step size $\alpha^{(k)} > 0$ using line search (e.g. Armijo or Wolfe)
    Update iterate: $x^{(k+1)} \leftarrow x^{(k)} + \alpha^{(k)} p^{(k)}$
    Compute new gradient: $g^{(k+1)} \leftarrow \nabla f(x^{(k+1)})$
    Compute step: $\delta^{(k)} \leftarrow x^{(k+1)} - x^{(k)}$
    Compute gradient difference: $y^{(k)} \leftarrow g^{(k+1)} - g^{(k)}$
    if $\left(y^{(k)}\right)^T \delta^{(k)} > 0$ **then**
        $$\rho^{(k)} \leftarrow \frac{1}{\left(y^{(k)}\right)^T \delta^{(k)}}$$
        $$H^{(k+1)} \leftarrow \left(I - \rho^{(k)} \delta^{(k)} \left(y^{(k)}\right)^T\right) H^{(k)} \left(I - \rho^{(k)} y^{(k)} \left(\delta^{(k)}\right)^T\right) + \rho^{(k)} \delta^{(k)} \left(\delta^{(k)}\right)^T$$
    else
        Skip update: $H^{(k+1)} \leftarrow H^{(k)}$
    end if
    $k \leftarrow k + 1$
end while
return $x^{(k)}$

Remarks on the BFGS Algorithm

- BFGS updates an **inverse Hessian approximation** $H^{(k)}$ using only gradient information.
- Starting with
$$H^{(0)} = I$$
gives a symmetric positive definite initial matrix.
- If
$$\left(y^{(k)}\right)^T \delta^{(k)} > 0,$$
then positive definiteness is preserved.
- Therefore,
$$p^{(k)} = -H^{(k)} g^{(k)}$$
remains a descent direction.

Why BFGS is Often Better than Gradient Descent

Gradient descent uses:

$$p^{(k)} = -g^{(k)}.$$

BFGS uses:

$$p^{(k)} = -H^{(k)} g^{(k)},$$

where $H^{(k)}$ approximates inverse curvature.
So BFGS can:

- scale directions differently
- reduce zig-zagging
- follow narrow valleys more efficiently
- converge much faster near the optimum

Near a minimizer, BFGS often behaves similarly to Newton's method, but without explicitly computing the true Hessian.

Applications of BFGS

BFGS is especially useful for:

- small to medium-sized optimization problems
- models requiring high precision
- smooth and differentiable regression / classification problems
- statistical parameter estimation (e.g. GLMs, maximum likelihood)
- smooth optimization in Bayesian inference
- training Hidden Markov Models
- hyper-parameter tuning
- some reinforcement learning optimization tasks

Limitations of BFGS

BFGS has important limitations:

- storing $H^{(k)} \in \mathbb{R}^{n \times n}$ requires $O(n^2)$ memory
- each update becomes expensive for very large n
- full BFGS is often impractical for deep learning-scale models
- requires smooth differentiability

For large-scale problems, a common alternative is:

L-BFGS (Limited-memory BFGS)

which stores only a few recent secant pairs.

Final Summary

- The secant pair

$$\delta^{(k)} = x^{(k+1)} - x^{(k)}, \qquad y^{(k)} = \nabla f(x^{(k+1)}) - \nabla f(x^{(k)})$$

  captures local curvature information.
- The secant equations are:

$$B^{(k+1)}\delta^{(k)} = y^{(k)}, \qquad H^{(k+1)}y^{(k)} = \delta^{(k)}.$$

- BFGS updates the inverse Hessian approximation while preserving symmetry and, under the curvature condition, positive definiteness.
- This gives a stable, curvature-aware, and often much faster method than plain gradient descent.

Numerical Example: Positive Curvature ($\left(y^{(k)}\right)^T \delta^{(k)} > 0$)

Consider the convex quadratic function

$$f(x_1, x_2) = \frac{1}{2}(x_1^2 + 2x_2^2).$$

Its gradient is

$$\nabla f(x) = \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}.$$

Choose two iterates:

$$x^{(k-1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad x^{(k)} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Then the step is

$$\delta^{(k)} = x^{(k)} - x^{(k-1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Numerical Example: Computing $y^{(k)}$

Compute the gradients:

$$\nabla f(x^{(k-1)}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \nabla f(x^{(k)}) = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

So,

$$y^{(k)} = \nabla f(x^{(k)}) - \nabla f(x^{(k-1)}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Now compute the curvature quantity:

$$\left(y^{(k)}\right)^T \delta^{(k)} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 > 0.$$

Conclusion: The curvature condition holds, so

$$\rho^{(k)} = \frac{1}{\left(y^{(k)}\right)^T \delta^{(k)}} = \frac{1}{3} > 0.$$

Hence, BFGS can safely update the inverse Hessian approximation.

Numerical Illustration of the Curvature Condition

Case 1: Positive curvature

$$\delta^{(k)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \left(y^{(k)}\right)^T \delta^{(k)} = 3 > 0$$

BFGS update is valid and $\rho^{(k)} > 0$.

Case 2: Zero curvature

$$\delta^{(k)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \left(y^{(k)}\right)^T \delta^{(k)} = 0$$

No detectable curvature; update is usually skipped.

Case 3: Negative curvature

$$\delta^{(k)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \left(y^{(k)}\right)^T \delta^{(k)} = -2 < 0$$

Numerical Example: One-Step BFGS Inverse Hessian Update

Given

$$H^{(k)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \delta^{(k)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

First check the curvature condition:

$$(y^{(k)})^T \delta^{(k)} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5 > 0 \quad \Rightarrow \quad \rho^{(k)} = \frac{1}{5}.$$

Using the BFGS inverse update formula,

$$H^{(k+1)} = \left(I - \rho^{(k)} \delta^{(k)} (y^{(k)})^T \right) H^{(k)} \left(I - \rho^{(k)} y^{(k)} (\delta^{(k)})^T \right) + \rho^{(k)} \delta^{(k)} (\delta^{(k)})^T,$$

we obtain

$$\boxed{H^{(k+1)} = \begin{bmatrix} \frac{23}{25} & -\frac{7}{25} \\ -\frac{7}{25} & \frac{13}{25} \end{bmatrix}}$$

This updated matrix is symmetric and incorporates curvature information from the secant pair.

Example: One BFGS Update

- **Question: Given** $H^{(k)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\delta^{(k)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $y^{(k)} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

  $x^{(k)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \nabla f(x^{(k)}) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \alpha^{(k)} = 1$. Using one BFGS iteration,

  1. compute the updated inverse Hessian $H^{(k+1)}$,
  2. compute the search direction $p^{(k)}$,
  3. determine the next iterate $x^{(k+1)}$.
- Recall

  $$H^{(k+1)} = \left(I - \rho^{(k)}\delta^{(k)}(y^{(k)})^T\right) H^{(k)} \left(I - \rho^{(k)}y^{(k)}(\delta^{(k)})^T\right) + \rho^{(k)}\delta^{(k)}(\delta^{(k)})^T,$$

  where $\rho^{(k)} = \frac{1}{(y^{(k)})^T\delta^{(k)}}$.
- First compute the curvature quantity:

  $$(y^{(k)})^T\delta^{(k)} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5 > 0.$$

  It means the curvature condition is satisfied.
- Therefore, $\rho^{(k)} = \frac{1}{(y^{(k)})^T\delta^{(k)}} = \boxed{\frac{1}{5}}$.

BFGS Update: Step 2 – First Matrix

- Compute

$$\delta^{(k)}(y^{(k)})^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}.$$

- Hence,

$$I - \rho^{(k)}\delta^{(k)}(y^{(k)})^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{5}\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}.$$

- Thus,

$$I - \rho^{(k)}\delta^{(k)}(y^{(k)})^T = \boxed{\begin{bmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{2}{5} & \frac{2}{5} \end{bmatrix}}.$$

- Compute

$$y^{(k)}(\delta^{(k)})^T = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}.$$

- Therefore,

$$I - \rho^{(k)}y^{(k)}(\delta^{(k)})^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{5}\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}.$$

- Hence,

$$I - \rho^{(k)} y^{(k)} \left(\delta^{(k)}\right)^T = \boxed{\begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix}}.$$

- Compute

$$\delta^{(k)} \left(\delta^{(k)}\right)^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- Therefore,

$$\rho^{(k)} \delta^{(k)} \left(\delta^{(k)}\right)^T = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- Substitute all terms into the BFGS update:

$$H^{(k+1)} = \begin{bmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{2}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- Since $H^{(k)} = I$, $H^{(k+1)} = \begin{bmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{2}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
- The matrix product is

$$\begin{bmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{2}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{18}{25} & -\frac{12}{25} \\ -\frac{12}{25} & \frac{8}{25} \end{bmatrix}.$$

BFGS Update: Final Result

- Add the rank-one term:

$$H^{(k+1)} = \begin{bmatrix} \frac{18}{25} & -\frac{12}{25} \\ -\frac{12}{25} & \frac{8}{25} \end{bmatrix} + \begin{bmatrix} \frac{5}{25} & \frac{5}{25} \\ \frac{5}{25} & \frac{5}{25} \end{bmatrix}.$$

- Therefore,

$$\boxed{H^{(k+1)} = \begin{bmatrix} \frac{23}{25} & -\frac{7}{25} \\ -\frac{7}{25} & \frac{13}{25} \end{bmatrix}}.$$

- Thus, one BFGS iteration transforms

$$H^{(k)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \longrightarrow \quad \boxed{H^{(k+1)} = \begin{bmatrix} \frac{23}{25} & -\frac{7}{25} \\ -\frac{7}{25} & \frac{13}{25} \end{bmatrix}}.$$

- The BFGS search direction is

$$p^{(k)} = -H^{(k+1)} \nabla f(x^{(k)}).$$

- Since the current gradient is given as $\nabla f(x^{(k)}) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

- Then

$$p^{(k)} = -\begin{bmatrix} \frac{23}{25} & -\frac{7}{25} \\ -\frac{7}{25} & \frac{13}{25} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

$$= -\begin{bmatrix} \frac{46}{25} - \frac{21}{25} \\ -\frac{14}{25} + \frac{39}{25} \end{bmatrix} = -\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- The next iterate is obtained by

$$x^{(k+1)} = x^{(k)} + \alpha^{(k)} p^{(k)}.$$

- Using

$$x^{(k)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \alpha^{(k)} = 1, \qquad p^{(k)} = \begin{bmatrix} -1 \\ -1 \end{bmatrix},$$

we get

$$x^{(k+1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

- Hence,

$$\boxed{x^{(k+1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}.$$

Question):
For each secant pair $(\delta^{(k)}, y^{(k)})$, compute $(y^{(k)})^T \delta^{(k)}$ and determine whether the BFGS update should be performed or skipped. Briefly justify your answer.

$$\text{(i)} \quad \delta^{(k)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{(ii)} \quad \delta^{(k)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\text{(iii)} \quad \delta^{(k)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y^{(k)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Question:

Consider one iteration of the BFGS method with

$$H^{(k)} = I, \qquad \delta^{(k)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y^{(k)} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

(a) Verify whether the curvature condition for BFGS is satisfied.

(b) Compute $\rho^{(k)}$.

(c) Explain whether the BFGS update should be performed or skipped.

(d) Without computing the full matrix $H^{(k+1)}$, explain what properties the updated inverse Hessian approximation will preserve if the update is performed.